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Product vision: The Employee Management System is an application designed to help companies manage their employees in an easier, more intuitive, and secure way. The application allows users to create, update, and delete employee information, departments, salary details, and leave records. It also provides downloadable reports of recent activities, salary expenses, and departmental activities.

With this application, companies can reduce the time spent on manual searches and data handling, improving organization and overall efficiency.

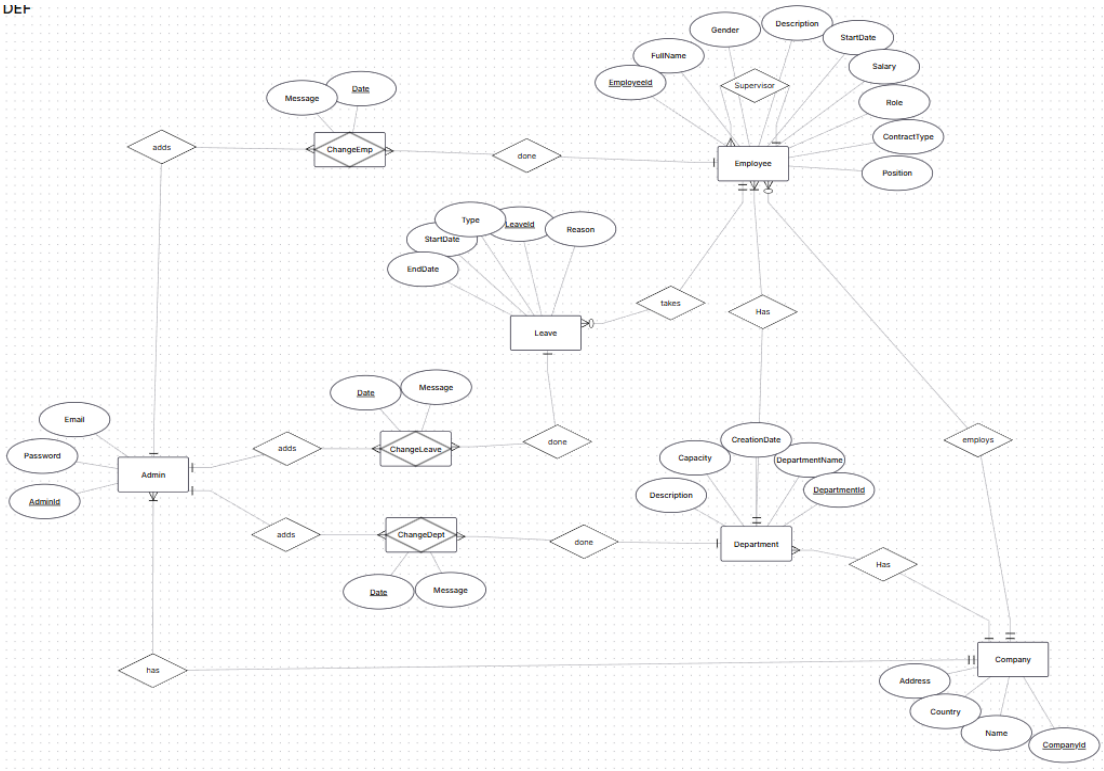


Figure 1: ER Diagram of Employee Management App

Entities

- **Company**: Saves the companies detail information such as name, address and country.
- **Admin**: Represents system administrators who manage the system. Admins are responsible for making changes to employees, leaves, and departments.
- **Employee**: Represents employees working in the company. Stores employee's information like name, salary or role.

- **Department**: Represents the team that each employee is belonging to. It has attributes such us capacity, description and name.
- **Leave**: Represents leave records taken by employees. Tracks the reason, type, and dates.
- **ChangeEmp (Associative Entity)**: Represents a record of changes made to employee data, saving date and message.
- **ChangeDept (Associative Entity)**: Represents change logs for department records, saving date and message.
- **ChangeLeave (Associative Entity)**: Represents change logs for leave records, saving date and message.

Relationships

- **Adds:** An admin adds (creates or updates) change records. One to many relationship.
  - Admin <-> ChangeEmp
  - Admin <-> ChangeLeave
  - Admin <-> ChangeDept
- **Done**: A specific change record is done on a particular entity. One to many relationship.
  - Department <-> ChangeDept
  - Employee <-> ChangeEmp
  - Leave <-> ChangeLeave
- **Takes:** An employee takes leave. One to many relationship.
  - Employee <-> Leave:
- **Has (Deparment-Employee):** A deparment has employees. Each employee belongs to one department. One to Many relationship.
  - Department <-> Employee
- **Has (Company-Department):** A company has departments. Each department belongs to one company. One to many relationship.
  - Company <-> Department
- **Has (Company-Admin):** A company has admins. Each adming belings to one company. One to many relationship.
  - Company <-> Admin
- **Employs**: A company employs employees. One to many relationship.
  - Company <-> Employee
- **Supervisor (Recursive Relationship):** One employee supevises other employees. An employee may have one supervisor, and a supervisor may manage multiple employees. One to many relationship.
  - Employee <-> Supervisor

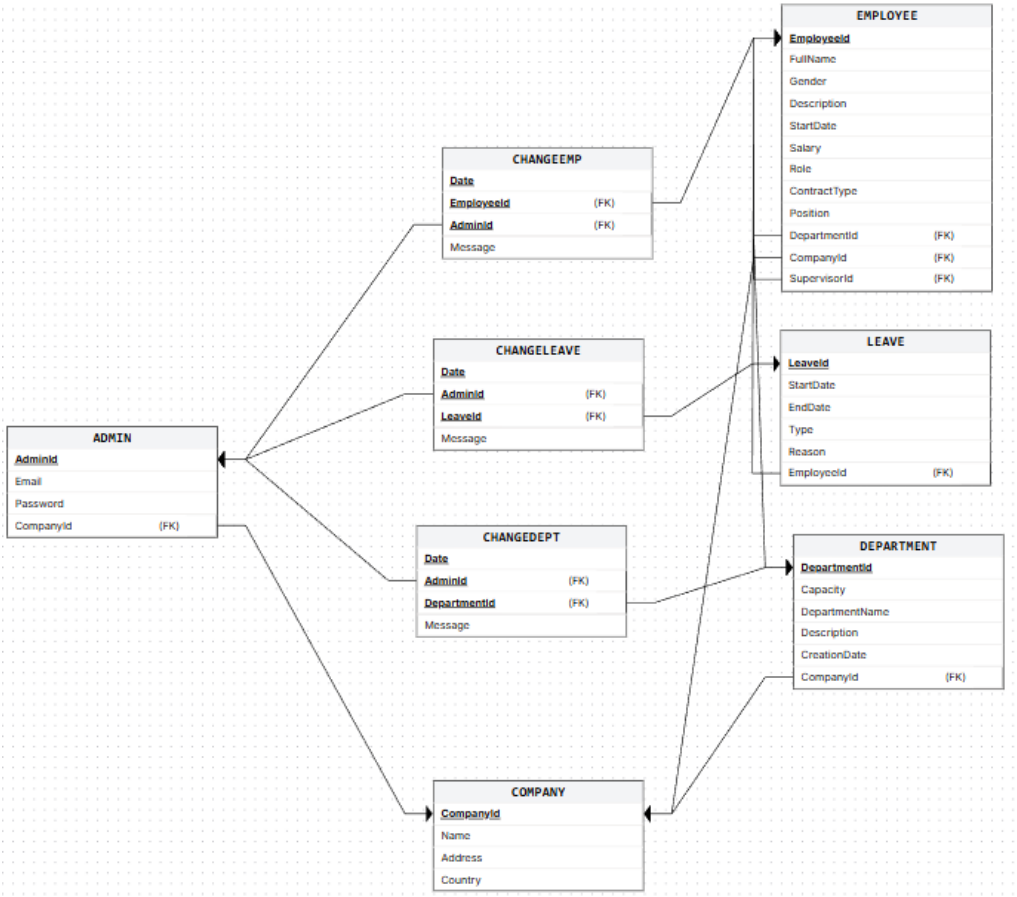


Figure 2: Relational database schema of Employee Management App

```
MariaDB [employment_management]> SELECT * FROM Company;
+------------+-------------------+---------+--------------+
| company_id | address           | country | name         |
+------------+-------------------+---------+--------------+
|          1 | 123 Tech Street   | USA     | TechCorp     |
|          2 | 456 Business Ave  | UK      | BizSolutions |
|          3 | Street 1          | Finland | TestCo       |
|          4 | Street 1          | Finland | TestCo       |
|          5 | Street 1          | Finland | TestCo       |
|          6 | Street 1          | Finland | TestCo       |
|          7 | Street 1          | Finland | TestCo       |
|          8 | Street 1          | Finland | TestCo       |
|          9 | Street 1          | Finland | TestCo       |
|         10 | Street 1          | Finland | TestCo       |
|         11 | Street 1          | Finland | TestCo       |
|         12 | Test Road 5       | Finland | JUnitCo      |
|         14 | Street 1          | Finland | NewName      |
|         15 | Street 1          | Finland | TestCo       |
|         16 | Street 1          | Finland | TestCo       |
```

Figure 3: Sample data of Company´s table.

```
MariaDB [employment_management]> SELECT * FROM Admin;
+----------+-----------------------+----------+------------+
| admin_id | email                 | password | company_id |
+----------+-----------------------+----------+------------+
|        1 | alice@techcorp.com    | pass123  |          1 |
|        2 | bob@bizsolutions.com  | pass123  |          2 |
|        3 | alice@test.com        | pass123  |          3 |
|        4 | alice@test.com        | pass123  |          4 |
|        5 | alice@test.com        | pass123  |          5 |
|        6 | alice@test.com        | pass123  |         15 |
|        8 | updated@test.com      | pass123  |         17 |
|        9 | alice@test.com        | pass123  |         18 |
|       10 | alice@test.com        | pass123  |         19 |
```

Figure 4: Sample data of Admin's table.

```
MariaDB [employment_management]> SELECT * FROM Employee;
+-------------+--------------+-------------+--------------+--------+-------------------+-----------+--------+------------+------------+---------------+
| employee_id | contractType | description | fullName     | gender | position          | role      | salary | startDate  | company_id | department_id |
+-------------+--------------+-------------+--------------+--------+-------------------+-----------+--------+------------+------------+---------------+
|           1 | Full-time    | Developer   | John Doe     | Male   | Software Engineer | Developer |   5000 | 2022-01-10 |          1 |             1 |
|           2 | Part-time    | HR Manager  | Jane Roe     | Female | HR Specialist     | HR Manager|   4500 | 2023-03-05 |          2 |             2 |
|           3 | Full-time    | Test employee | John Doe   | Male   | Engineer          | Developer |   3000 | 2024-01-01 |          6 |             6 |
|           5 | Full-time    | Test employee | Updated Name | Male | Engineer          | Developer |   3000 | 2024-01-01 |          8 |             8 |
|           6 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |          9 |             9 |
|           7 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         10 |            10 |
|           8 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         11 |            11 |
|           9 | Full-time    | Test employee | John Doe   | Male   | Engineer          | Developer |   3000 | 2024-01-01 |         21 |            15 |
|          11 | Full-time    | Test employee | Updated Name | Male | Engineer          | Developer |   3000 | 2024-01-01 |         23 |            17 |
|          12 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         24 |            18 |
|          13 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         25 |            19 |
|          14 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         26 |            20 |
|          15 | Full-time    | Test employee | John Doe   | Male   | Engineer          | Developer |   3000 | 2024-01-01 |         36 |            24 |
|          17 | Full-time    | Test employee | Updated Name | Male | Engineer          | Developer |   3000 | 2024-01-01 |         38 |            26 |
|          18 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         39 |            27 |
|          19 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         40 |            28 |
|          20 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         41 |            29 |
|          21 | Full-time    | Developer   | John Doe     | Male   | Software Engineer | Developer |   5000 | 2022-01-10 |         48 |            30 |
|          22 | Part-time    | HR Manager  | Jane Roe     | Female | HR Specialist     | HR Manager|   4500 | 2023-03-05 |         49 |            31 |
|          23 | Full-time    | Test employee | John Doe   | Male   | Engineer          | Developer |   3000 | 2024-01-01 |         53 |            35 |
|          25 | Full-time    | Test employee | Updated Name | Male | Engineer          | Developer |   3000 | 2024-01-01 |         55 |            37 |
|          26 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         56 |            38 |
|          27 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         57 |            39 |
|          28 | Full-time    | Test Employee | Jane Doe   | Female | Engineer          | Developer |   3000 | 2024-01-01 |         58 |            40 |
+-------------+--------------+-------------+--------------+--------+-------------------+-----------+--------+------------+------------+---------------+
```

Figure 5: Sample data of Employee's table.

```
MariaDB [employment_management]> SELECT * FROM Department;
+---------------+----------+--------------+-----------------+---------------------+------------+
| department_id | capacity | creationDate | departmentName  | description         | company_id |
+---------------+----------+--------------+-----------------+---------------------+------------+
|             1 |       10 | 2023-01-01   | IT              | Handles IT systems  |          1 |
|             2 |       15 | 2023-02-01   | HR              | Handles HR tasks    |          2 |
|             3 |       10 | 2026-02-10   | IT              | Tech department     |          3 |
|             5 |       10 | 2026-02-10   | Updated Dept    | Tech department     |          5 |
|             6 |       10 | 2026-02-10   | IT              | Tech dept           |          6 |
|             7 |       10 | 2026-02-10   | IT              | Tech dept           |          7 |
|             8 |       10 | 2026-02-10   | IT              | Tech dept           |          8 |
|             9 |       10 | 2026-02-10   | IT              | Tech dept           |          9 |
|            10 |       10 | 2026-02-10   | IT              | Tech dept           |         10 |
|            11 |       10 | 2026-02-10   | IT              | Tech dept           |         11 |
|            12 |       10 | 2026-02-10   | IT              | Tech department     |         18 |
|            14 |       10 | 2026-02-10   | Updated Dept    | Tech department     |         20 |
|            15 |       10 | 2026-02-10   | IT              | Tech dept           |         21 |
|            16 |       10 | 2026-02-10   | IT              | Tech dept           |         22 |
|            17 |       10 | 2026-02-10   | IT              | Tech dept           |         23 |
```

Figure 6: Sample of Department's table.

```
MariaDB [employment_management]> SELECT * FROM employee_leave;
+----------+------------+----------------+------------+------------------+-------------+
| leave_id | endDate    | reason         | startDate  | type             | employee_id |
+----------+------------+----------------+------------+------------------+-------------+
|        1 | 2024-01-06 | Family trip    | 2024-01-01 | Vacation         |           1 |
|        2 | 2024-01-12 | Flu            | 2024-01-10 | Sick Leave       |           6 |
|        4 | 2024-02-05 | Updated reason | 2024-02-01 | Updated Vacation |           8 |
|        5 | 2024-01-12 | Flu            | 2024-01-10 | Sick Leave       |          12 |
|        7 | 2024-02-05 | Updated reason | 2024-02-01 | Updated Vacation |          14 |
|        8 | 2024-01-12 | Flu            | 2024-01-10 | Sick Leave       |          18 |
|       10 | 2024-02-05 | Updated reason | 2024-02-01 | Updated Vacation |          20 |
|       11 | 2024-01-06 | Family trip    | 2024-01-01 | Vacation         |          21 |
|       12 | 2024-01-12 | Flu            | 2024-01-10 | Sick Leave       |          26 |
|       14 | 2024-02-05 | Updated reason | 2024-02-01 | Updated Vacation |          28 |
+----------+------------+----------------+------------+------------------+-------------+
```

Figure 7: Sample data of Leave's table.

```
MariaDB [employment_management]> SELECT * FROM change_emp;
+----------------------------+--------------------------------+----------+-------------+
| date                       | message                        | admin_id | employee_id |
+----------------------------+--------------------------------+----------+-------------+
| 2026-02-09 13:18:37.620937 | Updated employee salary to 5000 |       29 |          29 |
+----------------------------+--------------------------------+----------+-------------+
1 row in set (0.000 sec)
```

Figure 8: Sample data of ChangeEmp's table.

```
MariaDB [employment_management]> SELECT * FROM change_leave;
+----------------------------+-------------------------+----------+----------+
| date                       | message                 | admin_id | leave_id |
+----------------------------+-------------------------+----------+----------+
| 2026-02-11 13:18:37.621944 | Leave approved by admin |       29 |       15 |
+----------------------------+-------------------------+----------+----------+
1 row in set (0.000 sec)
```

Figure 9: Sample data of ChangeLeave's table.

```
MariaDB [employment_management]> SELECT * FROM change_dept;
+----------------------------+-------------------------------+----------+---------------+
| date                       | message                       | admin_id | department_id |
+----------------------------+-------------------------------+----------+---------------+
| 2026-02-10 13:18:37.621944 | Department description updated |       29 |            41 |
+----------------------------+-------------------------------+----------+---------------+
1 row in set (0.000 sec)
```

Figure 10: Sample data of ChangeDept table.